

PVX-4130 $\pm 6,000V$ Pulse Generator



The PVX-4130 pulse generator produces fast, high voltage wave forms to 6,000V. Optimized for high impedance capacitive loads, the PVX-4130 is well suited for driving extraction grids and deflection plates for electrostatic modulation of particle beams in time-of-flight mass spectrometers and accelerators. Its robust and versatile design also makes it well suited for pulsing or gating power tube grids, Pockels cells and Q Switches, acoustic transducers, microchannel plates, photomultiplier tubes and image intensifiers. The exceptional pulse fidelity of the PVX-4130 will optimize the performance of any system in which it is used.

The PVX-4130 generates an output voltage pulse of 6,000 volts with rise and fall times less than 60ns, with very flat voltage pulses to DC into a capacitive load. It can generate singled-ended output pulses from ground to +6000V or from ground to -6000V, and can also generate pulses originating from a DC voltage offset from ground by using both VLow and VHigh power supply inputs. This offset can be from -6000V to +6000V, with a maximum power supply voltage differential of $\leq 6000V$.

The PVX-4130 requires a TTL gate signal, a high voltage DC power supply and optional DC offset supply inputs. The output pulse width and frequency are controlled by the gate signal. The pulse output voltage is controlled by the amplitude of the input DC power supplies.

When the input gate is high, the V_{high} supply is connected to the output. When the input gate is low, the V_{low} supply is connected to the output. Therefore the PVX-4130 can be used to generate a negative-going pulse by logically inverting the input gate, so that the input gate is high until the unit is pulsed. When the input gate goes low, the V_{low} input supply is connected to the output, thereby generating a negative-going pulse.

- **0 to $\pm 6000V$ Pulse Output**
- **<60ns Rise and Fall Times**
- **<150ns to DC Pulse Width**
- **>10KHz Pulse Repetition Frequency**
- **Optimized to Drive Deflection Plates, Grids, and other Capacitive Loads**
- **Protected Against Arcs, Shorts, and Load Transients**
- **Voltage and Current Monitor Outputs**

The PVX-4130 features front panel indicator LEDs to monitor the status of the pulse generator. Front panel voltage and current monitors provide a straightforward means to view the output voltage and current waveforms in real-time, eliminating the need for an external high voltage oscilloscope probe.

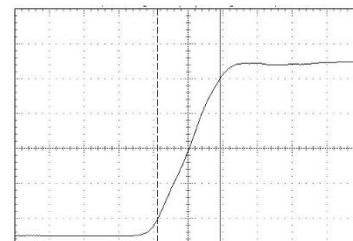
The pulse generator is a direct-coupled, air-cooled solid-state half-bridge (totem pole) design, offering equally fast pulse rise and fall times, low power dissipation, and virtually no over-shoot, under-shoot or ringing. It has over-current detection and shut-down circuitry to protect the pulse generator from potential damage due to arcs and shorts in the load or interconnect cable. All control and protection logic circuitry, support power, energy storage and output network are incorporated into the PVX-4130. It can be connected directly to the load, and does not require series or shunt resistors, impedance-matching networks between the pulser and the load, or additional energy storage (capacitor banks). All of this is taken care of within the PVX-4130.

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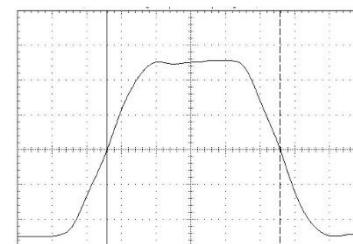


Specifications All specifications measured into a 50pF load connected w/4 ft (~1.2m) of RF-58 (50Ω) coaxial cable

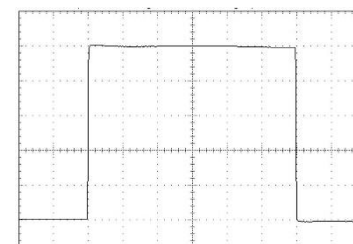
OUTPUT	
Maximum Value	± 6000 Volts ($V_{high} - V_{low}$)
Minimum Value	0 Volts
Means of Adjustment	Controlled by Power Supply Input Voltages
Pulse Rise and Fall Time	<60ns, typically <52ns (10% to 90%)
Pulse Width	<150ns to DC, Controlled by Input Gate
Pulse Recurrence Frequency (PRF)	Single shot to 10KHz at 6000V continuous output, maximum limited by power dissipation ⁽¹⁾
Max. Average Power	100W ($V_{high} - V_{low}$), derated at 2W/ °C over 25 °C ambient ⁽¹⁾
Max. Duty Cycle	Continuous
Droop	<1%
Over/undershoot	<5%
Throughout Delay	160ns Typical
Jitter	<1ns shot-to-shot
Output Connector & Cable	Kings 10KV, Rear Panel, with 4 feet (~1.2M) RG-58 (50Ω) Coaxial Cable
INPUT DC VOLTAGE $+V_{in}$ (V_{high}) (User-Supplied)	
Absolute Max. Value	+6000 Volts
Absolute Min. Value	-6000 Volts
Relative Max. Value	+6000 Volts over V_{low} Voltage
Relative Min. Value	+0V Over V_{low} Voltage
INPUT DC VOLTAGE $-V_{in}$ (V_{low}) (User-Supplied, only needed if biasing output)	
Absolute Max. Value	+6000 Volts
Absolute Min. Value	-6000 Volts
Input DC Connectors	Kings 10KV, Rear Panel (One each for $+V_{in}$ and $-V_{in}$)
GATE (User-Supplied)	
Gate Source & Connector	TTL into 50Ω, into BNC connector on the front panel
VOLTAGE & CURRENT MONITORS	
Voltage Monitor	2000: 1 into 1 MegΩ, BNC connector
Current Monitor	10A/V into 50Ω, BNC connector
GENERAL	
Support Power	90VAC to 240VAC, 50/60Hz
Dimensions (Excluding Connectors)	19"W x 5.2"H x 16"D (48.25cm W x 13.2cm H x 41cm D)
Weight (Approximate)	18 lbs. (8.2 Kilograms)
SPECIFICATIONS SUBJECT TO CHANGE WITHOUT NOTICE	



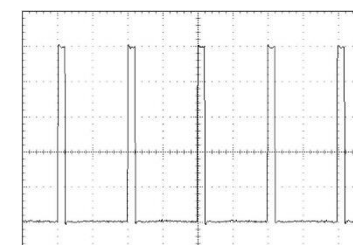
45ns Rise Time, 6000V Output
(25ns/Div horizontal scale, 1.2kV/Div vertical scale)



125ns Min. PWM, 6000V Output
(25ns/Div horizontal scale, 1.2kV/Div vertical scale)



Typical Output Waveform, 6000V Output
(25μs/Div horizontal scale, 1.2kV/Div vertical scale)



10KHz Repetition Frequency, 6000V Output
(50μs/Div horizontal scale, 1.2kV/Div vertical scale)

The specifications are measured driving a 50pF load connected with 4 feet of RG-58 cable, at 6000 V output. However, the PVX-4130 can drive loads of a few picofarads to several hundred picofarads of capacitance, limited by its maximum power dissipation capability ⁽¹⁾. At lower load capacitances and/or voltages less than 6000V, the PVX-4130 can also drive resistive or inductive loads, within limitations. Contact BNC for additional information and applications assistance.

⁽¹⁾ The power dissipated in the PVX-4130 when driving a capacitive load is defined by the formula CV^2F , where C is the total load capacitance, including the capacitance of the load, interconnect cable, and the internal capacitance of the PVX-4130, V is the pulse voltage, and F is the pulse repetition frequency (or the total pulses per second). (For these calculations, the internal capacitance of the PVX-4130 is 100pF, and RG-58 cable is 30pf/foot). Given the maximum dissipation of 100W, the maximum load capacitance, frequency and/or voltage at which the PVX-4130 can operate can be approximated using this formula. This formula also approximates the high voltage power supply requirements needed to drive a given load at a specific voltage and frequency. This formula is not applicable when driving resistive or inductive loads.